

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458252.1554 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.136 +/- 0.05
Transit depth (Rp/Rs)^2	1.86 +/- 1.36 [%]
Orbital Inclination (inc)	83.37 +/- 2.35
Airmass coefficient 1 (a1)	0.869 +/- 0.019
Airmass coefficient 2 (a2)	0.13 +/- 0.14
Scatter in the residuals of the lightcurve fit is	2.23 %
Best Comparison Star	#2 - [620, 128]
Optimal Aperture	3.27
Optimal Annulus	5.28
Transit Duration (day)	0.0004 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.136 ± 0.05 vs 0.1394 (deviation 0.07σ)
Duration fit vs published	0.0004 d vs 0.0983 d (deviation 15.48σ)
Mid-time O–C	-4030.8 min
Depth SNR	2.7
Residual scatter	2.23 %

Light curve

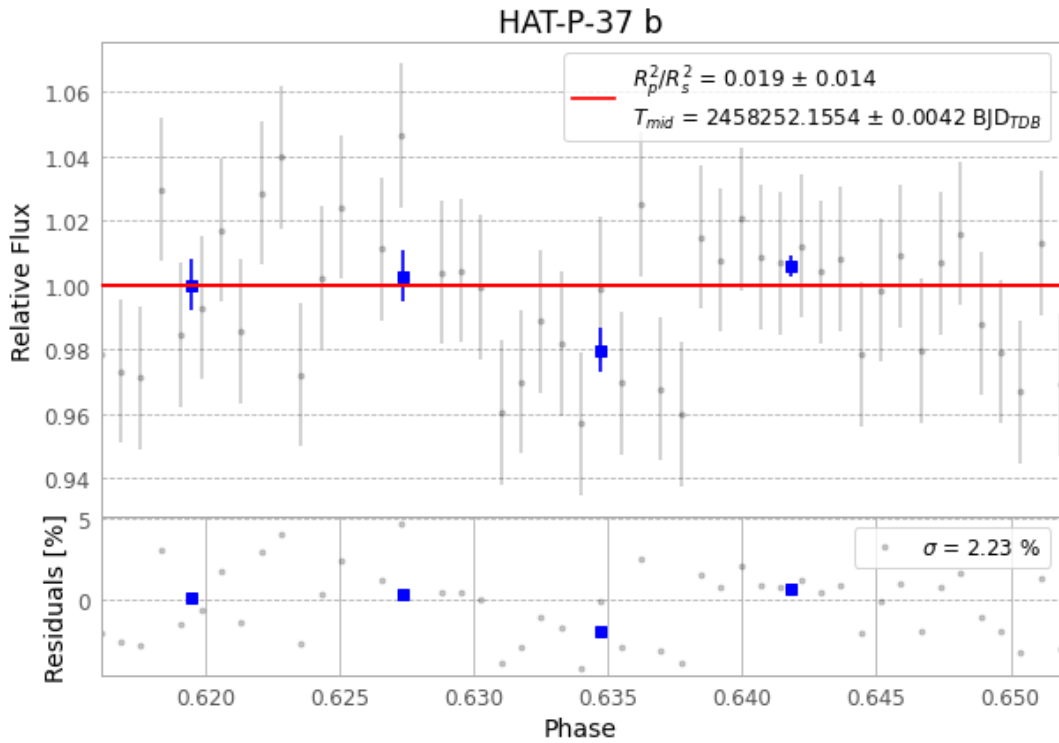


Figure 1 — FinalLightCurve_HAT-P-37 b_2018-05-15.png (SHA-256 08ac0dfe54a62350...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [376, 296], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b4ca444fe715deb6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

49 FITS frames (32 MB), HATP-37180515090312.FITS → HATP-37180515112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-180516.log (d6d5e5e8a7c1...), FinalLightCurve_HAT-P-37 b_2018-05-15.png (08ac0dfe54a6...), FinalLightCurve_HAT-P-37 b_2018-05-15.csv (989f0a4a7a83...)

Compute resources

Wall time 66.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 00:01Z.